

Difference-in-Differences

Design, Dynamics, Continuous and Staggered Treatment

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Causal inference and the counterfactual

Difference-in-differences (hereafter **DiD**) is a method of **causal inference** — it identifies a cause-and-effect relationship. It turns on one question: *what would have happened had the shock or policy never occurred?* From that **counterfactual** we recover the effect of the shock or the policy.

It is the very stuff of science fiction — stories built on an alternative history or a parallel world: *The Man in the High Castle* (P. K. Dick), *Back to the Future*, *Doctor Strange* (Marvel), and many more.



What is difference-in-differences?

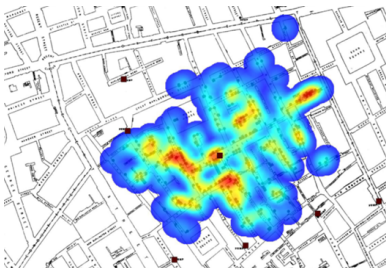
A policy hits **some units** at **some date**. We want its causal effect — but treated and untreated units differ systematically, so neither a treated-vs-control comparison nor a before-vs-after comparison is clean.

The idea in one line

Compare the **change** in the treated group to the **change** in a comparison group. The second difference nets out fixed level differences *and* common time shocks.

- Needs **panel** (or repeated cross-sections): the same groups, before and after.
- **Simple DiD is the TWFE model from the panel chapter** — the only new ingredient is one explanatory variable: a **dummy** for “treated group $\times$ after treatment”. We build from there to the modern extensions.

The origin: John Snow and cholera (1855)



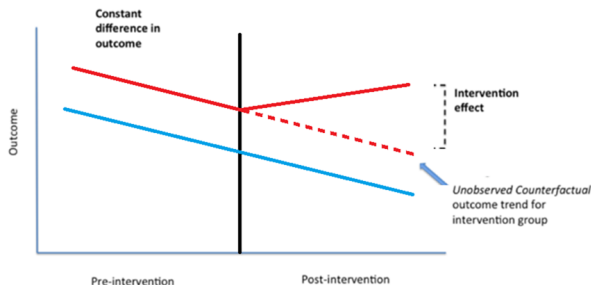
Origin: epidemiology — the fight against cholera in London. Cholera was assumed to be transmitted by **air**; John Snow (1855) argued instead that it came from **infected water**.

How to prove it? *A natural experiment.* The whole city drew water from the same point in the Thames, until a reform split the supply: some districts now received water from **downstream** of the sewers (infected), others from **upstream** (assumed uncontaminated).

⇒ *treated vs control, before vs after* — the first difference-in-differences.

The difference-in-differences logic

The **treatment** is that some neighbourhoods now receive **uncontaminated water**. We look at the **before vs after** for treated and untreated neighbourhoods (the **first difference**), then take the difference **between the two groups** (the **second difference**). Any factor that hits **both** groups at once is thereby **eliminated**.



Roadmap

- ➊ The **canonical** 2×2 — potential outcomes, parallel trends, the estimator.
- ➋ **Dynamics** — event studies and two-way fixed effects.
- ➌ **Staggered** adoption — why naive TWFE breaks, and the modern estimators.
- ➍ **Continuous** treatment — a “dose”, and the same heterogeneity lesson.

*Recent synthesis: Roth, Sant’Anna, Bilinski & Poe (2023). Textbook treatments: Cunningham, *Causal Inference: The Mixtape*; Békés and Kézdi (2021).*

Section 1

The canonical 2×2 design

Potential outcomes

Imagine two parallel worlds for each unit i :

- $Y_i(1)$ — the outcome it **would have if treated**;
- $Y_i(0)$ — the outcome it **would have if not treated**.

A **treatment indicator** $D_i \in \{0, 1\}$ ($1 = \text{treated}$) selects which one we actually see:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

The fundamental problem of causal inference

For any unit we observe **only one** potential outcome; the other — the **counterfactual** — is never seen.

The treatment effect, and the missing counterfactual

The causal effect for unit i is the gap between its two worlds:

$$\tau_i = Y_i(1) - Y_i(0).$$

A treated unit ($D_i = 1$) reveals $Y_i(1)$, but its $Y_i(0)$ — what would have happened **without** treatment — is the **missing counterfactual**.

- Individual effects are hopeless, so we target **averages over groups**.
- Notation: $\mathbb{E}[\cdot \mid D_i = 1]$ reads “the average **among treated** units”;
 $\mathbb{E}[\cdot \mid D_i = 0]$ “the average **among untreated** units” — the bar “|” means “restricted to”.

Two averages: ATT and ATU

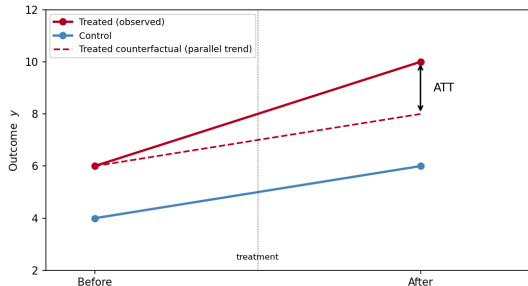
Average effect on the treated (ATT) and on the untreated (ATU)

$$ATT = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1], \quad ATU = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 0].$$

- **ATT** — the effect for those who **were** treated (“did the policy help its recipients?”). **ATU** — the effect it **would have** had on the untreated (“should we extend it?”). They differ when effects are heterogeneous.
- **DiD targets the ATT.** The one missing piece is the treated group’s untreated counterfactual $\mathbb{E}[Y_i(0) \mid D_i = 1]$ — which **parallel trends** will supply.

The two-by-two: two differences

Two groups, two periods — we now index outcomes by time, $Y_{it}(d)$. DiD builds the treated group's missing counterfactual by assuming it would have moved like the control:



$$\widehat{ATT} = \underbrace{(\bar{y}_{T,\text{post}} - \bar{y}_{T,\text{pre}})}_{\text{treated change}} - \underbrace{(\bar{y}_{C,\text{post}} - \bar{y}_{C,\text{pre}})}_{\text{control change}}$$

Parallel trends: the identifying assumption

Parallel trends

Had the treated group *not* been treated, its outcome would have changed over time by the **same amount** as the control group. Writing $\Delta Y_i(0)$ for that change in the **untreated** outcome:

$$\mathbb{E}[\Delta Y_i(0) \mid D_i = 1] = \mathbb{E}[\Delta Y_i(0) \mid D_i = 0].$$

- **Levels** may differ freely (fixed effects absorb them) — only the **trend** must match.
- **Untestable** (it concerns an unobserved counterfactual): defend it with pre-period evidence and economic argument — you cannot prove it.



The regression form

The 2×2 DiD is one OLS regression:

$$y_{it} = \alpha + \gamma D_i + \lambda \text{Post}_t + \beta (D_i \times \text{Post}_t) + \varepsilon_{it}.$$

- γ — the fixed treated-vs-control gap; λ — the common time shock.
- β — **the DiD coefficient** = $\widehat{\text{ATT}}$ under parallel trends; the interaction $D_i \times \text{Post}_t$ equals d_{it} (one for treated units in the post period, zero otherwise).
- With a panel this equals two-way fixed effects (next section).

Example: Card and Krueger (1994)

New Jersey raised its minimum wage (1992); neighbouring Pennsylvania did not. Compare fast-food employment across the state line.

FTE employment	Before	After	Change
New Jersey (treated)	20.44	21.03	+0.59
Pennsylvania (control)	23.33	21.17	-2.16
Difference-in-differences			+2.76

- Employment in New Jersey **rose** relative to Pennsylvania — not just “did not fall” — against the textbook competitive prediction.
- The cross-border design makes parallel trends plausible: similar labour markets, same macro shocks.

Assumptions and threats

Beyond parallel trends, a credible DiD needs:

- **No anticipation** — units don't react before treatment, or the pre-period is contaminated.
- **SUTVA** (stable unit treatment value) — two parts: (i) **no interference**, one unit's treatment does not change another's outcome (no spillovers across the border); (ii) **one version** of treatment, so "treated" means the same thing for every unit.
- **Stable composition** — no differential entry/exit (attrition) changing who is in each group.
- **Inference** — the treatment dummy is highly **serially correlated**, so naive standard errors badly understate uncertainty (the Bertrand–Duflo–Mullainathan 2004 problem from the panel chapter) $\Rightarrow$ **cluster** by unit/group.

Section 2

Dynamics: event studies

From the 2×2 to panel TWFE

With many units and periods, write the DiD as **two-way fixed effects**:

$$y_{it} = f_i + f_t + \beta d_{it} + u_{it}, \quad d_{it} = D_i \cdot \text{Post}_t.$$

- f_i absorbs fixed unit differences, f_t the common time shocks (recall the panel chapter). With **common timing** (all treated units adopt at the same date), $\beta = \widehat{\text{ATT}}$ — an equality that **breaks under staggered timing** (the staggered section).
- The static β hides two questions: do effects **grow over time**? do **pre-trends** hold? $\rightarrow$ the event study.

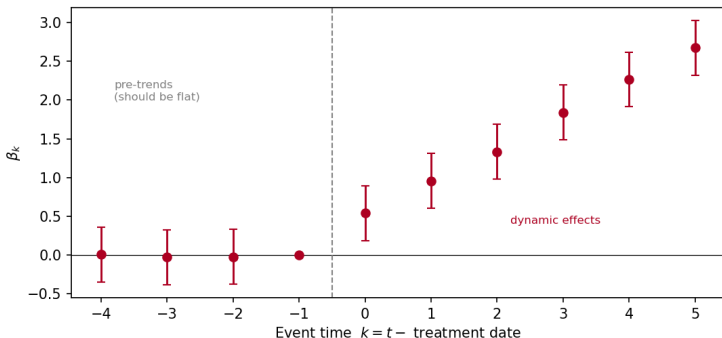
The event-study specification

Replace the single switch by a full set of **leads and lags** around each unit's treatment date:

$$y_{it} = f_i + f_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - t_i^* = k\} + u_{it}.$$

- $k = t - t_i^*$ is **event time** (t_i^* = unit i 's treatment date, called g_i in the staggered section); $k = -1$ is the **omitted baseline**, so each β_k is relative to the period just before treatment.
- β_k for $k < 0$ are **pre-trends**; for $k \geq 0$ they are **dynamic treatment effects**. (Bin the extreme k to avoid under-identified endpoints.)

Reading the event study



- **Flat, near-zero** coefficients before 0 are consistent with parallel trends; the post coefficients trace the effect's **dynamics**.
- A sloping pre-period ($\beta_k \neq 0$ for $k < 0$) is a red flag for the design.

A caution on pre-trends

Passing a pre-trend test is reassuring, not decisive

Tests of $\beta_k = 0$ for $k < 0$ often have **low power** — they miss real violations — and conditioning on "passing" can itself distort the estimate (Roth, 2022).

- Treat them as **evidence**, alongside an economic case for why the trends should align.
- **Plot** the coefficients; never lean on a single p -value (on event-study design and visualization, see Freyaldenhoven et al. 2021).

Section 3

Staggered adoption

Units adopt at different times

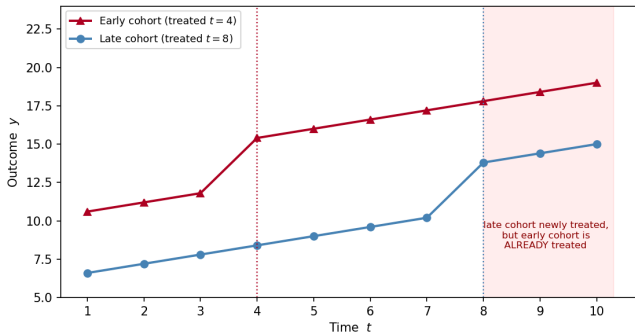
Most policies roll out **gradually**: unit i is first treated at date g_i .

- Example — Emission Control Areas: Baltic (2006), North Sea (2007), North America (2012), Caribbean (2014).
- The natural move is the same **static** TWFE — one switch $d_{it} = \mathbf{1}\{t \geq g_i\}$, one coefficient:

$$y_{it} = f_i + f_t + \beta d_{it} + u_{it}.$$

- The **dynamic** form you usually see swaps the switch for leads and lags, $\sum_{k \neq -1} \beta_k \mathbf{1}\{t - g_i = k\}$ — one coefficient per event time.
- **This is where it breaks** — the static β first (next slides), and the dynamic β_k with it under staggered timing.

The TWFE trap: forbidden comparisons



TWFE implicitly uses **already-treated** units as controls for **later-treated** ones — a **forbidden comparison**. The already-treated “control” is still responding to its own treatment, so its change is **not** a clean counterfactual.

Negative weights and sign reversal

TWFE under heterogeneity (Goodman-Bacon 2021; de Chaisemartin & D'Haultfœuille 2020)

The TWFE coefficient is a weighted average of all possible 2×2 DiDs — and some weights are **negative**. With heterogeneous effects, β can even take the **wrong sign** when every true effect is positive.

Goodman-Bacon (2021) makes this exact:

$$\hat{\beta}_{\text{TWFE}} = \sum_k w_k \widehat{\text{DiD}}_k, \quad \sum_k w_k = 1, \text{ some } w_k < 0,$$

the negative w_k sitting on the **forbidden** (already-treated-as-control) comparisons.

*A recurring theme: like the complier-weighting of **LATE** and the **Rotemberg weights** of shift-share (IV chapter), a regression coefficient is a weighted average of heterogeneous effects — here some weights turn negative, which is what breaks it.*

The building block: group-time effects

Modern estimators target the **group-time ATT** directly — the effect at time t for the cohort first treated at g :

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid g_i = g], \quad t \geq g.$$

- The missing $Y_{it}(0)$ is estimated from **clean** controls — never- or not-yet-treated units (no already-treated comparisons).
- **Aggregate** to one overall number (or an event study) with transparent, **non-negative** weights:

$$\text{ATT}^0 = \sum_{g,t} w(g, t) \text{ATT}(g, t), \quad w(g, t) \geq 0,$$

in contrast to the TWFE decomposition above, where some weights were *negative*.

The modern estimators

Estimator	Clean comparison group
Callaway & Sant'Anna (2021)	never-treated <i>or</i> not-yet-treated
Sun & Abraham (2021)	interaction-weighted; never-treated (or last-treated) cohort
Borusyak et al. (2024)	impute $Y(0)$ from not-yet-treated (efficient)
de Chaisemartin & D'Haultfœuille (2020)	consecutive periods; treatment can switch on <i>and off</i>

All replace the forbidden comparison with clean controls; they differ in the control set and in efficiency.

What you must compute

These estimators need the **timing** structure in the data:

- the **cohort** g_i — each unit's first treatment period (treated at $t = 6 \Rightarrow g_i = 6$ on *all* of its rows);
- **event time** $k = t - g_i$;
- a **never- / not-yet-treated** indicator, to define clean controls.

Software builds the comparisons for you — but you must supply g_i .

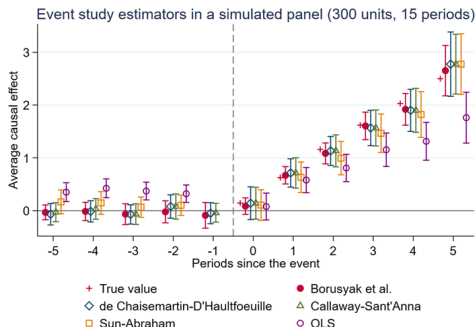
Example: unilateral divorce and suicide

Unilateral divorce and women's suicide (Goodman-Bacon 2021)

US states adopted unilateral divorce at staggered dates (California 1969 ...; a few never). Re-analysing Stevenson & Wolfers (2006), Goodman-Bacon shows the TWFE estimate of the fall in women's suicide is **smaller** than the timing-robust one (-3.1 vs -7.0).

- The already-treated comparisons **attenuate** the effect toward zero — here the bias shrinks the estimate rather than flipping its sign (both are possible).

Staggered event study: estimators compared



- **OLS/TWFE** (open circles) is **biased** as dynamic effects grow; the **modern estimators** track the truth.
- Even the **pre-period** coefficients move — each naive lead/lag is **contaminated** by other periods' effects (Sun & Abraham 2021).

Section 4

Continuous treatment

When treatment is a dose, not a switch

Often treatment has **intensity**, not just on/off:

- pollution exposure that **decays with distance** from the source;
- spending levels, tax rates, minimum-wage **levels**, regulatory **stringency**.

We then care about the effect of a **change in intensity** — not merely of treatment's existence.

The continuous DiD specification

With a continuous dose D_i switched on after the policy:

$$y_{it} = f_i + f_t + \beta (D_i \times \text{Post}_t) + u_{it}.$$

- β is a **dose-response slope**; the target is the **dose-specific** effect
 $\text{ATT}(d \mid d) = \mathbb{E}[Y_{it}(d) - Y_{it}(0) \mid D_i = d].$
- Identification needs a **stronger** parallel-trends assumption: units at *every* dose level would have trended alike absent treatment.

Callback — this *is* the shift-share design (IV chapter)

Exposure is a continuous dose; the Goldsmith-Pinkham et al. (2020) "shares view" assumption is parallel trends, and the estimator pools **many DiD comparisons — one per share.**

The catch: the same heterogeneity problem

Continuous TWFE is fragile (Callaway, Goodman-Bacon & Sant'Anna 2024)

If a given dose has **different effects across units**, the TWFE slope is a **non-convex weighted average** that mixes the dose-response with how units **select** into doses — it need not equal any dose effect, even under parallel trends.

- This is the **staggered problem again** — heterogeneous effects + implicit comparisons — now across **dose levels** rather than treatment **dates** (the dose analogue of shift-share's **Rotemberg-weight** diagnostics).
- Fix: estimate **dose-specific** effects and aggregate transparently (Callaway, Goodman-Bacon & Sant'Anna 2024; de Chaisemartin et al. 2023).

Section 5

Applied

DiD in Python (pyfixest)

```
import pyfixest as pf

# 1. Static TWFE (single date): beta = ATT
m = pf.feols("y ~ treatpost | id + year",
             df, vcov={"CRV1": "id"})

# 2. Event study: leads/lags, baseline k = -1
es = pf.feols("y ~ i(k, ref=-1) | id + year",
             df, vcov={"CRV1": "id"})

pf.iplot(es)           # plot beta_k with CIs

# 3. Staggered: do NOT trust plain TWFE -> use
#     pf.did2s (Gardner) or Callaway-Sant'Anna.
```

- `i(k, ref=-1)` builds the event-study leads/lags; `iplot` draws them.
- For **staggered** designs, always prefer a heterogeneity-robust estimator over raw TWFE.

Takeaways

- DiD turns **two differences** into a causal effect; identification is **parallel trends** — defended, not proved.
- **Event studies** show dynamics and probe pre-trends (low power $\Rightarrow$ argue, don't just test).
- **Continuous** and **staggered** designs break naive TWFE under **heterogeneous effects**: negative weights, forbidden comparisons, possible sign reversal.
- Use **group-time** estimators with clean controls (Callaway & Sant'Anna; Sun & Abraham; Borusyak et al.; de Chaisemartin & D'Haultfoeuille).

Next

Synthetic control — when no *single* clean control group exists, build a weighted one.

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